Auditing SHAP Explanations Against Latent Failure Mechanisms in Predictive Maintenance

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***Abstract*—Machine learning models predict equipment failures in industrial systems, but whether their explanations reflect the physical failure mechanisms is rarely tested statistically. We propose an audit that measures, for each known failure mechanism, how much SHAP attribution falls on the mechanism's trigger features, how specific that concentration is to the mechanism, and whether any other feature takes unusual attribution. Explanations are computed out of sample, and each statistic is compared with up to 1,000 models retrained on shuffled labels. On the AI4I 2020 dataset, a tuned XGBoost classifier (test F1 = 0.89) shows significant alignment for three of four physical mechanisms across five seeds, against a null re-tuned on every permutation, with a random forest, and with interventional TreeSHAP; a neural network explained by Integrated Gradients or KernelSHAP shows weaker alignment. The tool-wear verdict depends on the SHAP formulation, which we trace mainly to the reference distribution against which attributions are measured. A planted shortcut sensor shows that models can keep high accuracy while relying on a misleading feature, that weak shortcuts escape detection, and that at 300,000 observations only the shortcut statistic still detects a strong one. Predictive accuracy alone therefore cannot establish the physical relevance of explanations.**

***Index Terms*—Explainable AI, SHAP, predictive maintenance, feature attribution evaluation, physical failure mechanisms**

I. Introduction

Predictive maintenance uses machine-condition data to anticipate faults and schedule maintenance, reducing unplanned downtime [1]. Models range from decision trees to gradient-boosted ensembles and neural networks [2], [3], and the more complex ones are difficult to interpret [4]. Knowing which sensor readings drive a prediction helps maintenance teams assess an alarm [5], and explanation methods such as LIME [2], [6] and SHAP [5], [7] are now common in the field, including studies of how preprocessing and model complexity affect explanation quality [8].

High predictive performance can coexist with reliance on shortcuts or correlations unrelated to the intended task [9], [10]. In predictive maintenance, this raises the question of whether explanations emphasize the expected features for each failure type. SHAP assigns feature contributions to individual predictions [4], but these contributions do not by themselves establish the physical relevance of those inputs [11]. Evaluating feature attributions is difficult when a suitable ground-truth reference is unavailable [12]. Assessing mechanism alignment therefore requires information about which failure mechanism occurred and which features its rule depends on. Label-permutation methods provide a statistical reference for evaluating feature importance when the association between inputs and outcomes is disrupted [13]. The question is therefore not only which features receive attribution, but whether their attribution is consistent with the recorded mechanism and exceeds that observed under this reference.

In this work, we use the AI4I 2020 predictive maintenance dataset, a synthetic dataset containing machine operating measurements, an overall failure label, and separate failure-type labels [14]. Its recorded failure types and documented generation rules provide an external reference for evaluating mechanism alignment. Previous work on this dataset has assessed explanation quality using selected examples [15], while subsequent work has used SHAP to interpret feature importance [16]. More recently, Obeten and Jimoh have quantitatively compared SHAP and LIME feature rankings on AI4I 2020 [17]. Agreement between explanation methods, however, does not by itself establish alignment with a failure mechanism. We build on these studies by testing mechanism-level attribution on held-out observations against a reference obtained from models retrained on shuffled labels. We also examine whether the audit detects reliance on a deliberately fabricated feature.

Our audit trains a classifier on the overall failure label only and uses the recorded failure types to group its SHAP explanations by mechanism (Fig. 1). Three statistics measure the attribution share on a mechanism's expected trigger features, how much larger that share is than on other failures, and the largest excess share taken by any other feature. Explanations come from five-fold cross-fitting, so no observation is explained by a model trained on it, and each statistic is compared with pipelines retrained on shuffled labels, with Holm correction across mechanisms.

Our contributions are as follows:

- A mechanism-level SHAP audit with held-out explanations and a retrained shuffled-label null; we show that a random-subset baseline can approve a model trained on meaningless labels.
- A robustness study across seeds, features, a re-tuned null, two tree ensembles, three explainer families, and the ex-

plainer's background, locating the tool-wear disagreement in the reference distribution.
- A planted-shortcut study showing that high accuracy can conceal shortcut reliance and that weak shortcuts escape detection.
- A scaling study up to 300,000 observations (runtime up to one million), showing that only the shortcut statistic catches a strong shortcut at the largest size.
- A benchmark of seven classifiers with confidence intervals, and released code, models, and reproduction scripts.

II. Related Work and Background

Predictive maintenance studies use decision trees, random forests, and gradient-boosted trees [2], as well as support vector methods and neural networks [3]. Complex models require post-hoc explanation, and surveys review approaches ranging from inherently interpretable models to surrogate models, LIME, and SHAP [1], [18]. Local explanations examine individual predictions, whereas global summaries describe recurring patterns; in maintenance, explaining a single alarm and understanding a model's general reliance on sensor measurements are related but different tasks. Our audit sits between the two: it aggregates local attributions within each recorded failure mechanism.

Steurtewagen and Van den Poel [5] combine sensor and report data with an XGBoost model and use Shapley values to support the diagnosis of machinery breakdowns in the petrochemical industry, and Jalali et al. [3] compare LIME, SHAP, and attribute-wise explanations on the PHM08-CMAPSS data. Closest to our setting, Matzka [15] assessed explanation quality on AI4I 2020 against its known failure mechanisms using five examples from each of four physical failure types, a direct precedent for comparing explanations with the features expected to matter. Later work interpreted SHAP rankings on the same data [16], and Obeten and Jimoh measured SHAP–LIME agreement [17]. We extend this direction from selected examples and method agreement to a statistical test over all held-out observations.

SHAP assigns each feature an additive contribution that, together with a baseline value, reconstructs the explained output [4]. Covert et al. [19] organize removal-based explanations by how they handle unavailable features, which model behavior they explain, and how they summarize each feature's influence, showing that the assumption used to represent unavailable features is part of the explanation's definition. Shapley variants accordingly differ, conditioning on the remaining features, marginalizing over a background distribution, or substituting a single baseline [20], and recent surveys show that these choices answer different questions about the model and the data [21], [22], particularly when inputs are correlated [23], [24]. For tree ensembles, path-dependent TreeSHAP weights each path by its training cover, whereas interventional TreeSHAP averages over an explicit background sample [25]. Faster exact algorithms, such as Linear TreeSHAP [26] and the quadrature-based Quadrature-TreeSHAP [27], accelerate the path-dependent computation while retaining its formulation. The choice matters here because engineered features share information with the measurements they are built from, and Section V-D shows that it can decide a mechanism's verdict; agreement with trigger features must therefore be read in light of the explanation formulation, not as proof of physical causation.

Explanation-evaluation research offers complementary checks. Benchmarks with known ground truth, such as XAI-Bench [28], BAM [29], and CLEVR-XAI [30], test whether attributions recover planted relevance; OpenXAI evaluates faithfulness, stability, and fairness [31]; and ROAR [32] and its retraining-free successor ROAD [33] remove highly attributed inputs and measure the change in performance. Global sensitivity analysis based on the Hilbert–Schmidt independence criterion (HSIC) measures statistical dependence between inputs and outputs without a removal step [34]. These approaches characterize the model or its explainer; assessing agreement with a failure mechanism additionally requires an external reference, such as the documented failure rules that AI4I 2020 provides.

Randomization provides the reference we use. Label-permutation tests assess classifier performance [35], and outcome permutations calibrate feature-importance measures [13]. Some visually convincing saliency methods are insensitive to randomized parameters or labels [36] or unreliable under simple input transformations [37], and post-hoc explanations such as LIME and SHAP can be deliberately manipulated [38], so explanations need explicit controls. Research on shortcut learning gives a further reason: models can exploit predictive associations that do not represent the intended mechanism [9], [10].

Together, these studies motivate testing both whether explanations agree with known failure mechanisms and whether they reveal unexpected feature reliance. We address both with a mechanism-level audit of held-out SHAP attributions against documented trigger features, tested against models retrained on shuffled labels with correction for multiple comparisons, and with a planted-shortcut experiment that measures what the audit detects and what it misses.

III. Methodology

A. Problem Definition

A model may predict a failure correctly without relying on the conditions that caused it. We therefore ask: *when a particular failure occurs, does the model attribute more importance to the features in that failure's physical rule?*

Let $\mathcal{D} = \{(x_i, y_i, z_i)\}_{i=1}^{N}$ denote a dataset where $x_i \in \mathbb{R}^d$ contains the input features, $y_i \in \{0, 1\}$ indicates overall equipment failure, and $z_i \in \{0, 1\}^M$ records M individual failure mechanisms. A classifier f is trained using only (x_i, y_i); the mechanism indicators are reserved for evaluating its explanations.

For each mechanism m, domain knowledge identifies a set E_m of *trigger features* involved in its physical rule. We define $S_m = \{i : z_{i,m} = 1\}$ as the observations where mechanism m occurs and $F = \{i : y_i = 1\}$ as all observations labeled as

TABLE I
AI4I 2020 FAILURE MECHANISMS [15], THEIR TRIGGER CONDITIONS, EXPECTED FEATURES E_m, AND NUMBER OF FLAGGED OBSERVATIONS n. L/M/H ARE PRODUCT QUALITY TYPES.

Mech.	Trigger	E_m	n
TWF	tool wear at a random point in 200–240 min	tool wear	46
HDF	(process − air temp.) < 8.6 K and speed < 1380 rpm	both temps., their difference, speed	115
PWF	torque × speed (rad/s) outside [3500, 9000] W	torque, speed, power	95
OSF	tool wear × torque above 11,000/12,000/ 13,000 min·Nm for L/M/H	tool wear, torque, overstrain, its ratio	98
RNF	0.1% chance, independent of the process	–	19

failures. Let $\phi_j(x_i)$ denote the SHAP attribution of feature j for observation i.

The audit tests whether attribution concentrates on the expected trigger features, whether that concentration is specific to the mechanism, and whether unrelated features receive unusual attribution (Fig. 1). It measures agreement with known failure rules, not physical causation.

B. Dataset and Failure Mechanisms

AI4I 2020 [15] is a synthetic industrial predictive maintenance dataset representing a milling process. It contains 10,000 observations, each recording product quality (`L`, `M`, or `H`), air and process temperatures, rotational speed, torque, tool wear, an overall failure label, and five failure-mechanism indicators. Only 339 observations (3.4%) are labeled as failures, making the prediction task highly imbalanced.

The dataset describes five mechanisms: tool wear failure (TWF), heat dissipation failure (HDF), power failure (PWF), overstrain failure (OSF), and random failure (RNF). The overall failure label is intended to follow the logical OR of these mechanisms:

$$y_i^{\mathrm{OR}} = \mathbb{I}\left(\sum_{m=1}^{M} z_{i,m} \geq 1\right), \tag{1}$$

where $\mathbb{I}(\cdot)$ is the indicator function. Table I lists the documented rules and the trigger features used in the audit.

a) Overlapping mechanisms and label discrepancies: The released annotations do not always follow the intended OR relationship. Twenty-four observations have multiple active mechanism flags; these are valid overlapping conditions rather than labeling errors. However, nine observations are labeled as failures without any active flag, while 18 of the 19 RNF-flagged observations have an overall failure label of zero.

We train on the released overall labels. The primary audit includes every observation where the evaluated mechanism is active, and we repeat it on observations with exactly one active flag to remove ambiguity from overlaps. RNF has no trigger features and is examined only descriptively.

C. Feature Construction

Several mechanisms depend on combinations of sensor readings rather than individual measurements. We therefore construct four features from the original inputs: mechanical power $P = \tau(2\pi n/60)$, temperature difference $\Delta T = T_{\mathrm{process}} - T_{\mathrm{air}}$, overstrain $O = w\tau$, and overstrain ratio $Q = O/L(t)$, where τ, n, and w denote torque, rotational speed, and tool wear, respectively. The type-specific limit $L(t)$ is 11,000, 12,000, or 13,000 for product types `L`, `M`, and `H`.

They encode the interactions in the HDF, PWF, and OSF rules without using any labels, but they are correlated with the raw measurements, so we repeat the audit on the original features only (Section V-A).

D. Explanation Audit Statistics

We evaluate attribution using three complementary statistics. For feature j and observation set A, let

$$I_j(A) = \frac{1}{|A|}\sum_{i\in A}|\phi_j(x_i)|, \qquad s_j(A) = \frac{I_j(A)}{\sum_{k=1}^{d} I_k(A)} \tag{2}$$

denote the average absolute SHAP contribution and the attribution share of feature j, where $\phi_j(x_i)$ is computed by the cross-fitted model that did not train on observation i (Section III-E). We use absolute values because positive and negative contributions could otherwise cancel. The fraction of attribution assigned to the expected trigger features of mechanism m is

$$r_m(A) = \sum_{j\in E_m} s_j(A). \tag{3}$$

From these quantities, we define three statistics:

$$R_m = r_m(S_m), \tag{4}$$

$$D_m = r_m(S_m) - r_m(F \setminus S_m), \tag{5}$$

$$C_m = \max_{j\notin E_m}\left[s_j(S_m) - s_j(F\setminus S_m)\right], \tag{6}$$

R_m measures how much attribution goes to the trigger features on mechanism m's observations, D_m whether that share is larger than on other failures, and C_m the largest mechanism-specific increase among features outside E_m. A large C_m raises concern about reliance on another feature but does not prove the feature causally irrelevant.

For example, on the 95 PWF observations, power, torque, and speed have mean $|\phi|$ of 4.96, 1.93, and 1.33 (Fig. 3a), out of 11.82 over all twelve inputs, so $R_{\mathrm{PWF}} \approx 0.70$.

E. Out-of-Sample Explanations

Explanations of training observations may reflect overfitting, so we use five-fold cross-fitting [39]: the full pipeline is fitted on four stratified folds and explains the fifth, giving every observation an explanation from a model that did not train on it. Section V-A contrasts this with in-sample SHAP.

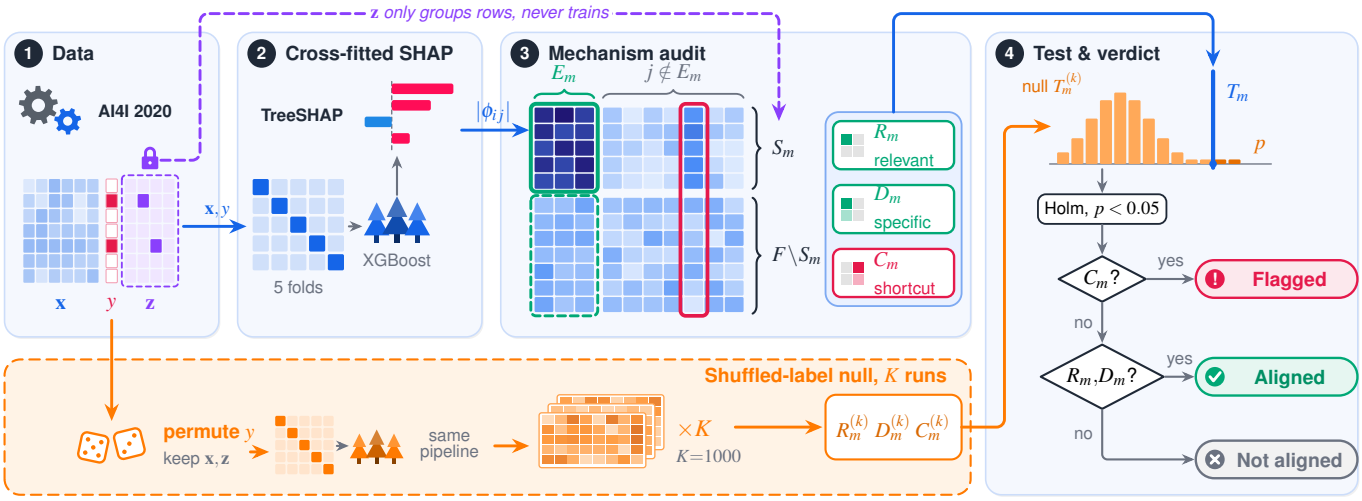


Fig. 1. Mechanism-level SHAP audit. (1) The classifier is trained on the sensor features $\mathbf{x}$ and the overall failure label y; the mechanism flags $\mathbf{z}$ are withheld from training and used only to group observations. (2) Five-fold cross-fitting gives every observation a TreeSHAP explanation from a model that did not train on it. (3) The absolute attributions $|\phi_{ij}|$ are split into mechanism observations S_m and other failures $F \setminus S_m$, and into trigger features E_m and all other features, giving the relevant share R_m, the specificity D_m, and the shortcut score C_m. Green marks the trigger features E_m (solid: S_m rows, giving R_m; solid versus dashed gives D_m); red marks the non-trigger feature with the largest excess, giving C_m. Cell values are illustrative. (4) Each observed statistic T_m is compared with its distribution over K shuffled-label runs (orange), in which y is permuted, $\mathbf{x}$ and $\mathbf{z}$ are kept, and the same pipeline is refitted. After Holm correction, a mechanism is flagged if C_m is significant, aligned if R_m and D_m are significant, and not aligned otherwise.

F. Statistical Significance

A high score alone is not evidence of alignment: mechanism observations often have extreme sensor values that receive large SHAP values even when the model has learned nothing, which makes a comparison against random failure observations unreliable (Section V-B).

We instead build a *shuffled-label null*: we permute the overall failure labels, keeping sensors and mechanism flags, and repeat the full preprocessing, oversampling, training, cross-fitting, and explanation procedure with the hyperparameters selected on the original data. Because carrying these hyperparameters over could itself shape the null, we also rebuild it with the full hyperparameter search repeated on every permutation (Section V-A).

For an observed value T and null values $T^{(1)}, \ldots, T^{(K)}$, the upper-tail Monte Carlo p-value is

$$p = \frac{1 + \sum_{k=1}^{K} \mathbb{I}(T^{(k)} \geq T)}{K + 1}. \tag{7}$$

The primary audit uses $K = 1{,}000$, giving a minimum possible p-value of $1/1001 \approx 0.001$. We apply Holm's correction [40] across the four physical mechanisms for each statistic and use $\alpha = 0.05$. To complement the p-values, we report a null-standardized effect size $z = (T - \bar{T}_0)/s_0$, where $\bar{T}_0$ and s_0 are the mean and standard deviation of the null values, and 95% percentile bootstrap intervals for R_m and D_m obtained by resampling observations within S_m and $F \setminus S_m$ (5,000 resamples). The bootstrap holds the cross-fitted models fixed, so it reflects observation sampling only; variability from refitting is captured by the null.

We combine the three tests into one decision rule. A mechanism is *aligned* when both R_m and D_m are significantly larger than their null distributions and C_m is not; it is *flagged* when C_m is significant, whatever R_m and D_m show, and *not aligned* otherwise. The shortcut test can therefore overturn a positive verdict, which matters at large n, where a narrow null lets R_m pass even when a non-trigger feature takes much of the attribution (Section V-E). A flag names the responsible feature and its excess share, which should be inspected before acting on it. A non-significant shortcut result means that the audit did not detect such an increase; it does not establish that shortcuts are absent. RNF is excluded from the physical-alignment tests because it has no expected trigger features.

G. Model Training and Evaluation

We split the data into stratified training, validation, and test sets (70/20/10), fitting scaling and one-hot encoding on the training data. To handle the 3.4% failure rate, we compare SMOTE [41], Borderline-SMOTE [42], and ADASYN [43] by the validation F1 of a default XGBoost [44]. We compare seven classifiers: logistic regression, RBF-kernel SVM, k-NN ($k = 7$), a decision tree (depth 10), a random forest (300 trees, depth 12), XGBoost (300 trees, depth 6, learning rate 0.1), and an MLP (64 and 32 hidden units), with library defaults otherwise (scikit-learn 1.9.1, XGBoost 3.4.1, PyTorch 2.14.0; seed 42). The best model, XGBoost, is tuned by a 40-iteration randomized search with five-fold cross-validation on F1 (depth 3–8, 200–600 trees, learning rate 0.01–0.2, subsampling 0.6–1.0, minimum child weight 1–5, γ 0–0.3), with oversampling inside each fold, and its threshold maximizes validation F1. We report failure-class precision, recall, F1, ROC-AUC, Brier score, and expected calibration error, with Clopper–Pearson

intervals for precision and recall, 5,000 stratified bootstrap resamples otherwise, and five-fold cross-validation at threshold 0.5.

H. Experimental Design

The primary audit uses $K = 1{,}000$ and takes about 12 minutes on eight CPU cores; variants use $K = 200$ unless stated, the shortcut sweep $K = 500$.

a) Robustness checks: We repeat the audit with a null re-tuned on every permutation, across four additional random seeds, on single-mechanism observations, without engineered features, and using in-sample rather than held-out SHAP values. We also compare path-dependent TreeSHAP with interventional TreeSHAP, using background samples of 100, 500, and 1,000 training observations per fold, drawn uniformly at random from that fold's original (not oversampled) training observations with up to three background seeds, and repeat the audit with a random forest. To examine the TWF result, we further draw the interventional background only from non-failures, only from failures, or from the oversampled training data, and run the interventional audit on the original features alone. Finally, to test whether the audit depends on tree ensembles or TreeSHAP, we audit the MLP of Section III-G with two non-tree explainers: KernelSHAP [4], a model-agnostic Shapley estimator (25 training observations as background, 1,000 coalitions per observation), and Integrated Gradients [45], a gradient-based method (baseline equal to the mean of 100 training observations, 1,024 steps, completeness checked to within 5%). Both explain the MLP's logit, and every null model retrains the MLP on permuted labels and is explained in the same way. Every variant uses a matching null.

b) Controlled shortcut experiment: To test whether the audit detects reliance on an unrelated but predictive feature, we introduce an artificial sensor

$$s_i = z_{i,\mathrm{HDF}} + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2). \tag{8}$$

This noisy copy of the HDF flag has no physical role and is excluded from E_{HDF}; we decrease σ from 2 to 0.1 to make it increasingly predictive.

c) Scaling experiment: We generate new observations using the documented AI4I 2020 process [15], without copying released rows. The sensor distributions follow the documented process, and HDF, PWF, and OSF use their published rules. Tool wear increases until a lifetime drawn from $U(200, 240)$ min, where TWF occurs with probability 0.4. The generated failure rate is 3.5–4.0%, compared with 3.4% in the release.

We evaluate $n \in \{10^4, 3 \cdot 10^4, 10^5, 3 \cdot 10^5\}$ with three generation seeds for the first three sizes and one for the largest. Each setting is tested without a shortcut and with shortcut noise $\sigma \in \{0.35, 0.25\}$. The scaling experiments use $K = 200$ null models, and the generator is included in the released code.

TABLE II
FAILURE-CLASS TEST PERFORMANCE, SORTED BY F1. PARENTHESIZED INTERVALS INDICATE 95% BOOTSTRAP CONFIDENCE INTERVALS FOR F1.

Model	Prec.	Rec.	F1 (95% CI)	ROC-AUC
XGBoost (tuned)	**1.00**	0.79	**0.89** (0.79–0.95)	0.988
XGBoost	0.86	0.88	0.87 (0.78–0.94)	0.988
Random forest	0.79	0.88	0.83 (0.74–0.92)	**0.989**
MLP	0.65	**0.91**	0.76 (0.67–0.84)	0.988
Decision tree	0.68	0.82	0.75 (0.64–0.84)	0.927
SVM	0.42	0.88	0.57 (0.48–0.65)	0.976
k-NN	0.43	0.79	0.56 (0.46–0.65)	0.877
Logistic regression	0.24	0.85	0.37 (0.31–0.43)	0.943

d) Why simulated data: AI4I 2020 documents its failure mechanisms, a reference rarely available in real data. The classifier sees only the overall label and may learn other relationships, so agreement with the rules is not automatic; the shuffled-label, shortcut, and raw-feature experiments test whether the audit distinguishes genuine from chance or shortcut agreement.

IV. Experimental Results

A. Predictive Performance

Borderline-SMOTE gives the highest validation F1 (0.826, versus 0.768 for SMOTE and 0.747 for ADASYN) and is used throughout. In Table II, tuned XGBoost reaches the highest test F1 (0.89 versus 0.87 untuned, threshold 0.978), although the confidence intervals overlap.

It finds 27 of the 34 test failures with no false alarm: precision 1.00 (95% CI 0.87–1.00), recall 0.79 (0.62–0.91), Brier score 0.0090 (0.033 for a base-rate predictor), and calibration error 0.008. Five-fold cross-validation gives failure-class F1 0.822 ± 0.013 and macro-F1 0.908 ± 0.007. This accuracy alone says nothing about whether the explanations match the failure mechanisms.

B. Alignment with Failure Mechanisms

Figure 2 and Table III compare the tuned XGBoost model's out-of-sample statistics with 1,000 shuffled-label models.

a) HDF, PWF, and OSF: All three mechanisms pass the relevant and specificity tests, with Holm-adjusted $p = 0.004$ for each. Their trigger features receive substantially more attribution on the corresponding mechanism observations than on other failures, with specificity scores between 0.33 and 0.43 and effect sizes of $z = 5.7$–8.4. The relevant-share effect is smallest for HDF ($z = 3.4$): its four trigger features include both temperatures and rotational speed, on which even shuffled-label models place much attribution (null mean 0.56). The bootstrap intervals are narrow (half-widths of 0.02–0.03) compared with the null standard deviations (0.04–0.06), so uncertainty in these verdicts comes mainly from model refitting rather than from the finite number of mechanism observations. No statistically significant shortcut is detected in the primary audit.

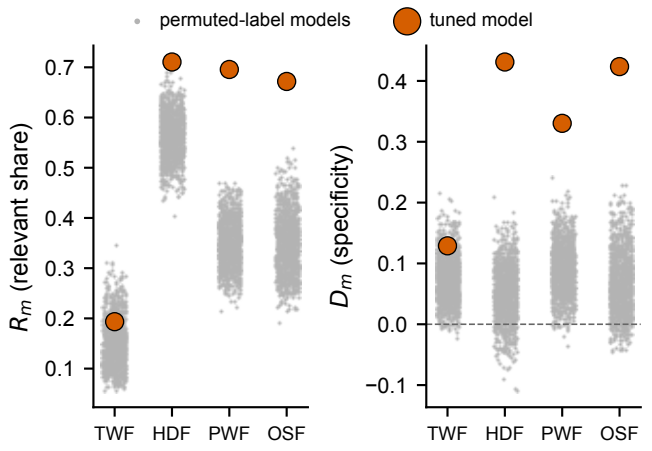


Fig. 2. Primary audit of the tuned XGBoost model. The observed relevant and specificity scores are compared with 1,000 shuffled-label models. HDF, PWF, and OSF exceed the null distributions on both measures, whereas TWF does not.

b) TWF: Tool wear failure does not pass either alignment test ($p_R = 0.201$, $p_D = 0.077$). Although tool wear is its most important feature, it receives only 19% of the attribution on TWF observations, compared with 15% under the null. The documented TWF mechanism includes a random failure point within a tool-wear interval, and only 46 observations carry this flag. Both factors may make its attribution harder to distinguish from the null.

C. Feature Attribution Across Mechanisms

In Fig. 3, the mechanism-level heatmap shows rotational speed and temperature difference dominating HDF, power and torque PWF, and the overstrain features OSF, whereas the global SHAP summary ranks features over all observations and hides these distinctions. A feature can be important to the classifier without explaining a given mechanism.

V. Experimental Analysis

Table IV summarizes how the audit responds to changes in the data, model, explainer, and null.

A. Robustness of Mechanism Alignment

a) Random seeds and overlapping mechanisms: HDF, PWF, and OSF pass both alignment tests across all five random seeds and when the analysis is restricted to observations with a single active mechanism. TWF fails both tests in every seed.

b) Re-tuned null: Repeating the 40-iteration hyperparameter search on each of 200 permutations gives a wider null than carrying over the hyperparameters selected on the real labels: the search on shuffled labels mostly selects shallow trees (depth at most 4 in 115 of 200 permutations), and the null 95th percentile of R_{OSF} rises from 0.46 to 0.62. The verdicts are unchanged. HDF, PWF, and OSF pass both tests and TWF fails both ($p_R = 0.453$, $p_D = 0.368$), but the OSF relevant-share margin narrows: 4 of 200 re-tuned null models reach the observed 0.672, giving an adjusted $p = 0.0498$. The carried-over null is therefore the less conservative reference, and the OSF relevant-share result should be read as borderline under the stricter one.

c) In-sample versus out-of-sample explanations: When explanations are calculated on observations used for training, all four mechanisms pass, including TWF ($p = 0.004$). Its relevant score increases from 0.19 with out-of-sample SHAP to 0.35 with in-sample SHAP. Explaining training observations thus gives an optimistic verdict.

d) Original features only: Removing the engineered features changes the verdict for HDF, which retains significant specificity but no longer passes the relevant-share test. PWF and OSF continue to pass both tests, although OSF specificity is close to the threshold ($p = 0.048$). TWF still fails. PWF and OSF alignment therefore does not depend on the derived features, whereas HDF's does, consistent with the difficulty of representing a temperature-difference rule with separate temperatures in a tree model.

B. Why the Choice of Null Matters

An audit should reject a model that learned nothing. A classifier trained on shuffled labels fails our alignment tests, but compared with random failure observations under the same model, all four mechanisms appear significant (adjusted $p = 0.002$): the naive comparison approves a model trained on meaningless labels, because mechanism observations have unusual sensor values that receive large attributions regardless. Retraining under shuffled labels is therefore necessary.

C. Can the Audit Detect a Shortcut?

Figure 4 shows attribution shifting from the HDF trigger features to the planted sensor as its noise decreases.

For weak shortcuts ($\sigma \geq 0.4$, at most 8% of the attribution), HDF passes. At 13% ($\sigma = 0.35$) the relevant-share test still passes ($p = 0.044$), and it fails from 21% ($\sigma = 0.30$). The shortcut statistic becomes significant at about 35% ($\sigma \leq 0.25$), and the specificity test fails at about 42%. Test F1 stays between 0.84 and 0.92 throughout, so the model keeps good accuracy while increasingly relying on an unrelated feature. PWF and OSF pass in every setting. Weak shortcuts thus go undetected, and the shortcut statistic is less sensitive than the relevant-share test at moderate strengths.

D. Effect of the Model and SHAP Method

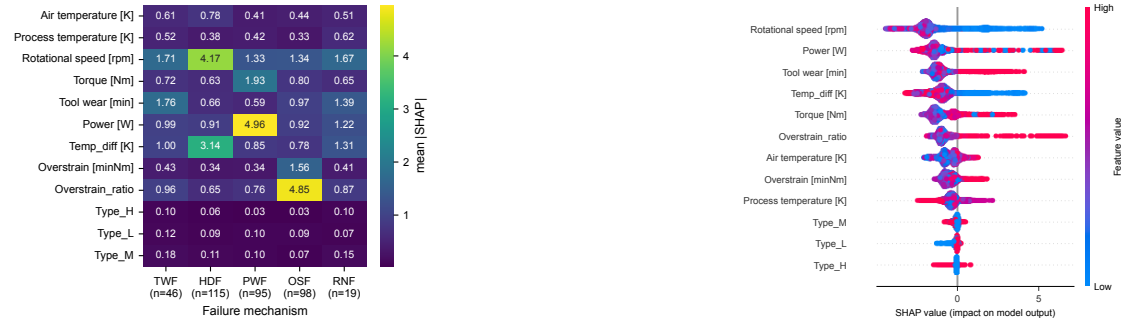
HDF, PWF, and OSF pass the alignment tests using both path-dependent and interventional TreeSHAP, and their verdicts remain unchanged with a random forest.

a) Beyond trees: The audit itself needs only an attribution matrix, so it runs unchanged on the MLP (test F1 0.76) with KernelSHAP and Integrated Gradients, each against 200 retrained MLP nulls (Table IV). With Integrated Gradients, HDF and OSF are aligned (adjusted $p \leq 0.030$ for both tests), PWF passes the relevant-share test narrowly ($p = 0.0498$) but not specificity ($p = 0.10$), and TWF fails. With KernelSHAP, no mechanism is aligned; only PWF specificity is significant. Two factors make these verdicts weaker than for XGBoost.

TABLE III
PRIMARY AUDIT USING CROSS-FITTED SHAP VALUES (SEED 42, $K = 1{,}000$).

	n	Top feature	R_m [95% CI]	Null	z	p	D_m [95% CI]	Null	z	p	C_m	Null	p
TWF	46	tool wear	0.193 [0.17, 0.22]	0.152	0.9	0.201$^{\times}$	0.129 [0.10, 0.16]	0.070	1.5	0.077$^{\times}$	0.025	0.025	1.000
HDF	115	speed	0.711 [0.69, 0.73]	0.557	3.4	0.004	0.431 [0.41, 0.45]	0.049	8.4	0.004	0.001	0.015	1.000
PWF	95	power	0.696 [0.67, 0.72]	0.347	7.6	0.004	0.330 [0.30, 0.36]	0.088	5.7	0.004	0.001	0.011	1.000
OSF	98	overstrain ratio	0.672 [0.66, 0.69]	0.345	5.6	0.004	0.424 [0.40, 0.44]	0.067	7.0	0.004	-0.001	0.019	1.000
RNF	19	speed	0.186	0.240	–	0.985	–	–	–	–	–	–	–

Note: Null: mean over the shuffled-label models; z: (observed − null mean)/null s.d.; brackets: 95% bootstrap intervals with the models held fixed; p: Holm-adjusted over the four physical mechanisms; $^{\times}$: non-significant alignment test.



(a) Mean absolute SHAP by failure mechanism

(b) Global SHAP summary

Fig. 3. SHAP explanations of the tuned classifier. (a) Attribution patterns for individual failure mechanisms. (b) Global feature importance across all observations. The mechanism-level view reveals distinctions that are not visible in the global ranking.

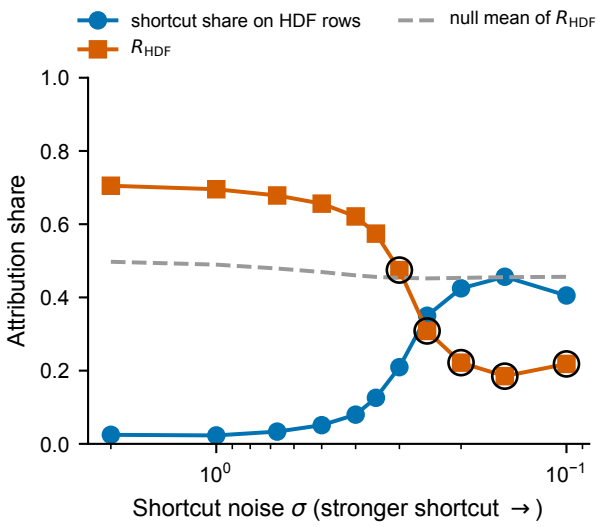


Fig. 4. Effect of an injected shortcut on HDF attribution. As noise decreases, the artificial sensor takes a larger share and the trigger features less. Markers indicate a failed relevant-share test.

First, shuffled-label MLPs already place much attribution on the trigger features, so the null means are high (e.g., R_{PWF} null mean 0.60–0.63 versus 0.35 for XGBoost), likely because a smooth network trained on noise still spreads attribution over the inputs that the PWF and OSF rules use. Second, the MLP is a weaker classifier, and KernelSHAP adds sampling error from its 25-row background and 1,000 of the 4,094 coalitions. We cannot separate these factors here, but the result shows why the shuffled-label null is needed: under KernelSHAP, the raw relevant shares of HDF, PWF, and OSF (0.47–0.67) look like alignment, yet none exceeds what an MLP without failure knowledge produces.

TWF behaves differently. It passes both tests with interventional TreeSHAP across the tested background sizes and seeds, with relevant scores between 0.45 and 0.47. However, it fails with path-dependent TreeSHAP and the random forest. Tool wear is strongly correlated with the engineered overstrain features overall (Spearman 0.92), and different SHAP formulations distribute attribution among dependent features differently [23]. Correlation alone does not explain the flip, however: on TWF observations the correlation is only 0.13, and path-dependent TWF alignment also fails without the engineered features (Table IV). Interventional TreeSHAP also passes TWF on the raw features alone ($R = 0.58$, $p = 0.020$).

The flip is better explained by the reference distribution against which each formulation measures a feature's contribution. Path-dependent TreeSHAP weights tree paths by the training cover, which reflects the oversampled data the model was fitted on, whereas our interventional background is drawn from the original training observations, about 97% of which are non-failures. We therefore repeat the interventional audit with three backgrounds (Table IV), each with a matching null. A non-failure background reproduces the default result ($R_{\mathrm{TWF}} = 0.46$). A background drawn from the oversampled training data lowers R_{TWF} to 0.23, close to the path-dependent 0.19, although TWF still passes narrowly ($p_R = 0.045$). A failure-only background lowers it to 0.08 and also breaks

TABLE IV
HOLM-ADJUSTED AUDIT p-VALUES ACROSS EXPERIMENTAL VARIANTS AND FAILURE MECHANISMS.

		TWF			HDF			PWF			OSF		
Variant	K	R	D	C	R	D	C	R	D	C	R	D	C
Primary, seed 42	1000	0.201$^{\times}$	0.077$^{\times}$	1.000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000
single-mechanism rows	1000	0.187$^{\times}$	0.076$^{\times}$	1.000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000
null re-tuned per permutation	200	0.453$^{\times}$	0.368$^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.0498	0.020	1.000
Seed 1	200	0.199$^{\times}$	0.119$^{\times}$	0.776	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Seed 2	200	0.149$^{\times}$	0.065$^{\times}$	0.498	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Seed 3	200	0.174$^{\times}$	0.095$^{\times}$	0.736	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Seed 4	200	0.269$^{\times}$	0.129$^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Raw features only	1000	0.345$^{\times}$	0.616$^{\times}$	1.000	0.190$^{\times}$	0.004	1.000	0.004	0.004	1.000	0.004	0.048	1.000
In-sample SHAP	1000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000	0.004	0.004	1.000
Interventional, background 100	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Interventional, background 500	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
Interventional, background 1000	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
failure-only background	200	0.995$^{\times}$	0.945$^{\times}$	0.020†	0.826$^{\times}$	0.149$^{\times}$	1.000	0.045	0.090$^{\times}$	1.000	0.040	0.020	1.000
non-failure background	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
oversampled background	200	0.045	0.020	0.896	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000
raw features only	200	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.020	0.020	0.975
Random forest	200	0.617$^{\times}$	0.398$^{\times}$	1.000	0.020	0.020	1.000	0.020	0.020	1.000	0.040	0.020	1.000
MLP, KernelSHAP	200	0.975$^{\times}$	0.831$^{\times}$	0.975	0.388$^{\times}$	0.189$^{\times}$	1.000	0.358$^{\times}$	0.020	1.000	0.358$^{\times}$	0.075$^{\times}$	1.000
MLP, Integrated Gradients	200	0.338$^{\times}$	0.100$^{\times}$	0.522	0.020	0.020	1.000	0.0498	0.100$^{\times}$	0.478	0.030	0.020	1.000
Permuted labels (sanity)	200	1.000$^{\times}$	0.731$^{\times}$	1.000	0.199$^{\times}$	0.557$^{\times}$	1.000	1.000$^{\times}$	0.731$^{\times}$	1.000	1.000$^{\times}$	0.731$^{\times}$	0.637
Naive null, tuned model	–	0.002	–	–	0.002	–	–	0.002	–	–	0.002	–	–
Naive null, permuted labels	–	0.002	–	–	0.002	–	–	0.002	–	–	0.002	–	–

Note: Holm-adjusted p-values of the R, D, and C tests; $^{\times}$: non-significant alignment test; †: significant shortcut test; "–": not evaluated. The minimum attainable adjusted p-value is $4/(K+1)$. Interventional results use background seed 0; other background seeds yield the same verdicts. Indented interventional rows use a background of 100 observations drawn from the stated training subset.

HDF alignment. The attribution therefore answers different questions under different references: against typical operation, tool wear is what distinguishes TWF observations, but against other failures it is not. Thus, the TWF verdict depends mainly on the choice of reference distribution rather than on feature correlation, and remains unresolved in the sense that no single choice is correct for all purposes. For maintenance audits that ask why an observation is abnormal, we recommend a background drawn from normal operation and reporting it alongside the result.

E. Does the Audit Work with More Data?

We repeat the audit on generated datasets containing 10^4 to $3 \cdot 10^5$ observations. Without an injected shortcut, HDF, PWF, and OSF pass the alignment tests at every evaluated size and seed. TWF continues to fail the relevant-share test, showing that additional observations alone do not resolve its alignment.

The shortcut experiments reveal a different pattern. At $\sigma = 0.25$, the artificial sensor receives 27–45% of HDF attribution. The relevant-share test detects the resulting loss of alignment in all nine runs up to 100,000 observations, but not at 300,000. At the largest size, the observed relevant score is 0.472, exceeding the null's 95th percentile of 0.417. As the null distribution becomes narrower, a model can remain significantly better than shuffled-label models while relying heavily on a misleading feature.

The shortcut score detects the stronger shortcut in all ten evaluated runs, including at 300,000 observations. At that size, its observed value is 0.249, compared with the null's 95th percentile of 0.211. Under the combined decision rule, HDF is therefore flagged rather than aligned in that run; across all experiments, this is the only case in which the shortcut test changes a verdict that R_m and D_m alone would have given. The weaker shortcut at $\sigma = 0.35$, which receives 6–20% of attribution, is not detected reliably.

The shortcut score also flags TWF in 7 of 30 runs from $3 \cdot 10^4$ observations onward, three of them without an injected sensor, always pointing to air or process temperature with small excess shares (0.03–0.05). The generator draws temperatures independently of tool life, so these flags cannot reflect a genuine shortcut. They likely arise because attribution on TWF observations, which the model captures poorly, is spread more evenly across features than on other failures, and the narrowing null at large n makes this small excess significant. A significant C_m should therefore be read together with the size of the excess share and the identity of the flagged feature.

a) Computational cost: The audit requires $5(K + 1)$ model fits because each real and randomized model is evaluated through five-fold cross-fitting. A single cross-fitted run takes approximately 1.3 seconds for 10^4 observations, 31 seconds for $3 \cdot 10^5$, and 115 seconds for 10^6 observations on one core, growing roughly linearly with n. With eight cores, the complete $K = 200$ audit takes about three minutes at 10^4 observations and approximately one hour at $3 \cdot 10^5$. The independent randomized runs can be distributed across multiple processors or machines.

TABLE V
AUDIT ON GENERATED DATA OF INCREASING SIZE ($K = 200$). SHARE: PLANTED SENSOR'S SHARE OF HDF ATTRIBUTION. COUNTS GIVE THE RUNS IN WHICH A TEST FIRES; MIN.: MINUTES PER AUDIT ON EIGHT CORES.

			HDF, planted sensor			TWF		
n	σ	Runs	Share	R fails	C flags	D passes	C flags	Min.
10,000	none	3	–	0/3	0/3	0/3	0/3	3
	0.35	3	0.158	2/3	0/3	0/3	0/3	3
	0.25	3	0.390	3/3	3/3	0/3	0/3	3
30,000	none	3	–	0/3	0/3	0/3	1/3	7
	0.35	3	0.128	0/3	0/3	2/3	0/3	6
	0.25	3	0.346	3/3	3/3	2/3	1/3	7
100,000	none	3	–	0/3	0/3	1/3	2/3	17
	0.35	3	0.100	0/3	0/3	2/3	1/3	19
	0.25	3	0.315	3/3	3/3	2/3	1/3	19
300,000	none	1	–	0/1	0/1	0/1	0/1	52
	0.35	1	0.063	0/1	0/1	0/1	0/1	59
	0.25	1	0.271	0/1	1/1	0/1	1/1	61

TABLE VI
SIMPLER AUDIT DESIGNS ON THREE CHECKS.

Design	Rejects shuffled labels	Avoids in-sample TWF pass	Detects shortcut at $3 \cdot 10^5$
Global SHAP ranking [16]	–	–	–
Naive null	No	No	No
In-sample SHAP	Yes	No	No
Relevant share R_m only	Yes	Yes	No
Full audit	**Yes**	**Yes**	**Yes**

"–": no statistical verdict. The naive null passes all four mechanisms ($p = 0.002$); in-sample SHAP, tested against the shuffled-label null, passes TWF ($p = 0.004$) and misses the shortcut ($p_C = 0.06$).

F. Comparison with Simpler Audit Designs

Table VI checks four simpler designs against three requirements: rejecting a shuffled-label model, avoiding the optimistic in-sample TWF verdict, and detecting the strong shortcut at 300,000 observations. Each misses at least one; the full audit meets all three, although it is not guaranteed to detect every misleading attribution.

VI. CONCLUSION

This study evaluated whether SHAP explanations of predictive maintenance models align with known physical failure mechanisms. Using AI4I 2020, we proposed a statistical audit that compares feature attributions with expected failure conditions and shuffled-label models. Despite achieving a test F1 of 0.89, the XGBoost model showed consistent alignment for three failure mechanisms, while tool wear results depended on the reference distribution used by the explanation method, and a neural network showed weaker, explainer-dependent alignment. The audit also revealed that accurate predictions can rely on misleading features and that simpler evaluation methods may fail to detect them. These findings highlight the need to evaluate explanations beyond predictive accuracy to better assess their agreement with known failure mechanisms.

For practitioners, our results suggest five rules. Explain only observations the model did not train on. Compare statistics with models retrained on shuffled labels, not with random subsets of failures. State the explainer's reference distribution, and for alarms ask why an observation is abnormal by drawing the background from normal operation. Report effect sizes and the flagged feature alongside p-values, and treat a mechanism as aligned only when R_m and D_m pass and C_m does not, since at large n a narrow null lets R_m pass despite a strong shortcut. Finally, re-tune the null models when the budget allows, as carried-over hyperparameters give a less conservative reference.

VII. LIMITATIONS AND FUTURE WORK

The audit is evaluated only on simulated AI4I 2020 data, so it establishes agreement with known simulator rules rather than verified physical causes, and it requires expert-defined trigger features. The TWF verdict depends on the explainer's reference distribution, and the MLP results depend on the explainer; LIME was not evaluated. The shortcut statistic misses shortcuts below about a third of the attribution, a studentized variant was less sensitive, and at large n small excesses can be flagged on poorly learned mechanisms. Under a null re-tuned on every permutation, OSF's relevant share is borderline ($p = 0.0498$), and the bootstrap intervals hold the models fixed. Future work will apply the audit to real data with recorded failure causes, such as the Tennessee Eastman Process [46], [47], develop more sensitive shortcut statistics, and add rank-based or distance-based statistics and dependence-based sensitivity indices [34] as complementary checks.

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